

Assessment Growth and Section Performance Analytics in R

Public-Safe BOY/EOY Improvement Study

Grant McCurdy

June 24, 2026

Recommendation

This study uses seven years of public-safe BOY/EOY assessment history to build an expected-score baseline for future instructional planning. The decision question is not which historical section requires action; those sections are already in the past. The decision question is which teachers and courses should receive closer review before the next assessment cycle because their historical growth patterns were directionally above or below expectation.

Use **BOY/EOY score gain** as the stakeholder metric: EOY score minus BOY score. Use the model only to set the expected baseline for students with similar starting scores, readiness, attendance, grade level, course track, and school-year context. A positive signal means growth was above expectation; a negative signal means growth was below expectation.

The future-facing review priorities are **TCH-001** for teacher support, **TCH-005** as a teacher bright spot, **Alg 1** for course support, and **Precalc** as a course bright spot. These are review priorities, not personnel ratings.

Priority	Target	Gap	Evidence
Teacher support	TCH-001	-0.17	23 historical sections and 229 records
Teacher bright spot	TCH-005	0.04	41 historical sections and 420 records
Course support	Alg 1	-0.34	10 historical sections and 93 records
Course bright spot	Precalc	+0.10	21 historical sections and 216 records

The gap column is the average observed-minus-expected BOY/EOY gain. The generated CSV also keeps the reliability-weighted review signal used to rank recurring patterns.

Approach and Rationale

The analysis includes 1,737 paired BOY/EOY records across 174 historical section-year groups and 5 simulated teachers. The mean BOY score is 48.5, the mean EOY score is 54.2, and the mean raw gain is 5.72 points.

The selected baseline is **EOY readiness model**, a Linear model using the Predicted EOY target. The model search tested 13 candidates across linear, interaction, GAM, random-forest, gradient-boosting, and leakage-check specifications.

The model's holdout expected-gain R-squared is **0.180**, while its holdout EOY R-squared is **0.947**. This difference is expected: BOY score explains much of final EOY score, but year-over-year gain is noisier after the starting score is removed. That is why the report aggregates signals at the teacher and course level instead of acting on individual predictions.

The operating baseline intentionally excludes teacher IDs, course IDs, and section IDs. Including those IDs would make the prediction slightly different, but it would also subtract away the persistent teacher and course patterns this study is designed to surface for future review.

Direct Answers

1. The main metric is BOY/EOY improvement: end-of-year score minus beginning-of-year score for the same public-safe student record.
2. The average raw gain is 5.72 points across 1,737 paired records.
3. The strongest baseline predicts EOY score and converts it to expected gain; holdout EOY R-squared is 0.947 and holdout expected-gain RMSE is 4.400.
4. Historical sections are evidence, not future action targets. The review layer flags 8 historical section-year groups above expected growth and 6 below expected growth, with 155 within expected range.
5. Recommended next-cycle review targets are TCH-001 for teacher support, TCH-005 for teacher bright-spot learning, Alg 1 for course support, and Precalc as a course bright spot.

Data Audit

The analysis starts from a public-safe assessment extract. A record enters the growth model only when the same public-safe student has valid BOY and EOY scores in the same section and with the same simulated teacher. This keeps the improvement metric tied to one section experience instead of mixing students across sections.

Measure	Value
Raw assessment rows	4,018
BOY/EOY candidate pairs	2,009
Included paired records	1,737
Unique public-safe student IDs	671
Unique section-year groups	174
Unique simulated teachers	5
Mean BOY score	48.5
Mean EOY score	54.2
Mean BOY/EOY gain	5.7
Median section paired records	10

Future Review Priorities

The recommendations below are the operational layer of the study. They use all seven years of historical evidence, weighted toward recurring teacher and course patterns. The purpose is to decide what to review before the next cycle, not to treat past sections as current operational units.

Priority	Target	Gap	Evidence
Teacher support	TCH-001	-0.17	23 historical sections and 229 records
Teacher bright spot	TCH-005	0.04	41 historical sections and 420 records
Course support	Alg 1	-0.34	10 historical sections and 93 records
Course bright spot	Precalc	+0.10	21 historical sections and 216 records

The table below shows the historical section evidence behind those priorities. It is intentionally placed after the future priorities because historical sections explain the signal; they are not the action target.

Priority	Section	Teacher	Course	N	Raw	Expected	Signal
T support	26-27 S03	TCH-001	Geometry	9	3.27	6.43	-1.50
T support	30-31 S01	TCH-001	Alg 1	10	3.28	6.10	-1.41
T bright	27-28 S17	TCH-005	AP Calc AB	11	8.79	5.74	1.60
T bright	30-31 S19	TCH-005	AP Calc AB	12	9.09	6.28	1.53
C support	30-31 S01	TCH-001	Alg 1	10	3.28	6.10	-1.41
C support	27-28 S01	TCH-001	Alg 1	10	5.83	7.38	-0.78
C bright	29-30 S17	TCH-003	Precalc	11	8.77	5.79	1.56
C bright	29-30 S18	TCH-004	Precalc	7	8.73	5.73	1.24

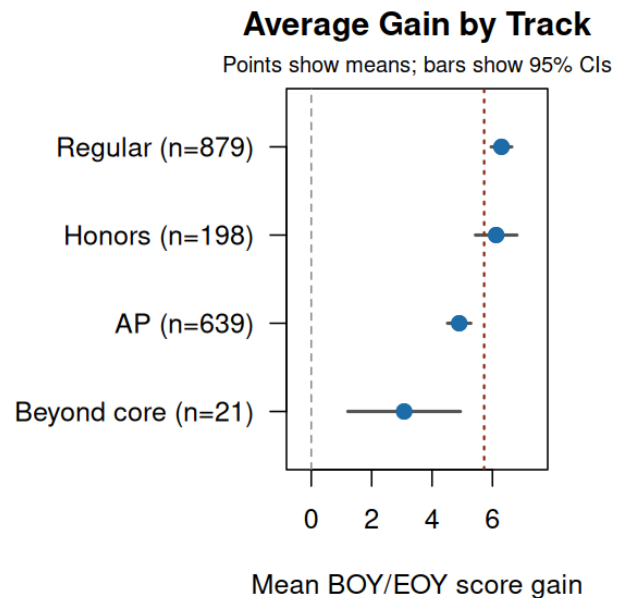
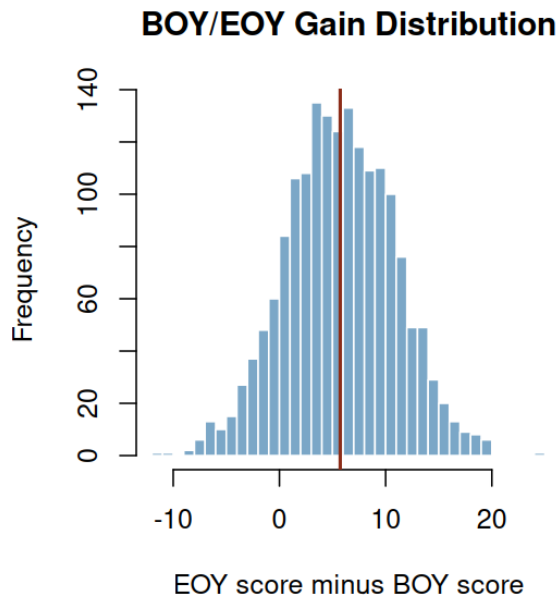
The full future-priority table is generated as [reports/future_review_priorities.csv](#), and the supporting historical evidence table is generated as [reports/historical_section_evidence.csv](#).

Raw Section Improvement

The first descriptive layer calculates BOY/EOY score gain inside each historical section-year group and runs a paired-improvement t-test against zero. This answers whether a section improved, but it does not by itself show whether the section improved more than expected for its starting profile.

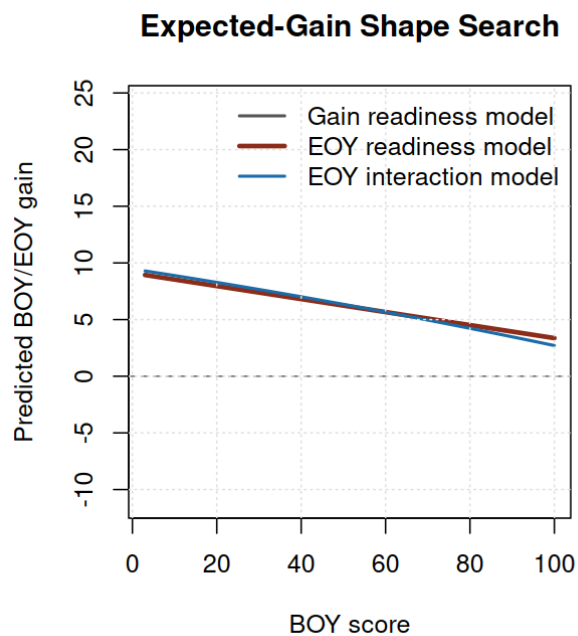
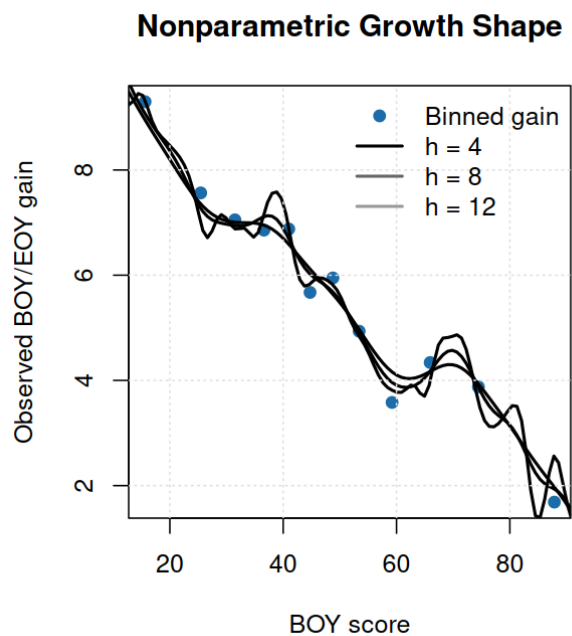
The table below shows high-signal historical section-year groups from the review layer, with raw BOY/EOY t-test results included for context. The full section t-test table is generated as [reports/section_ttests.csv](#).

Section	N	BOY	EOY	Gain	95% CI	p-value
30-31 S06	9	42.4	52.8	10.39	7.18 to 13.60	<0.001
27-28 S17	11	43.9	52.6	8.79	5.37 to 12.22	<0.001
29-30 S17	11	37.1	45.9	8.77	5.77 to 11.78	<0.001
30-31 S19	12	37.6	46.7	9.09	6.66 to 11.52	<0.001
29-30 S03	12	41.0	49.7	8.73	6.24 to 11.22	<0.001
31-32 S02	10	40.1	49.1	9.02	6.32 to 11.73	<0.001
29-30 S22	10	41.5	49.9	8.38	4.71 to 12.05	<0.001
30-31 S08	11	35.0	44.0	9.07	4.80 to 13.34	<0.001



Expected-Growth Baseline

The adjusted model estimates expected BOY/EOY gain from starting score/readiness and context. This is the key step that makes the analysis more useful than a raw gain ranking: it accounts for floor effects, ceiling effects, attendance context, course track, grade level, and school-year timing.



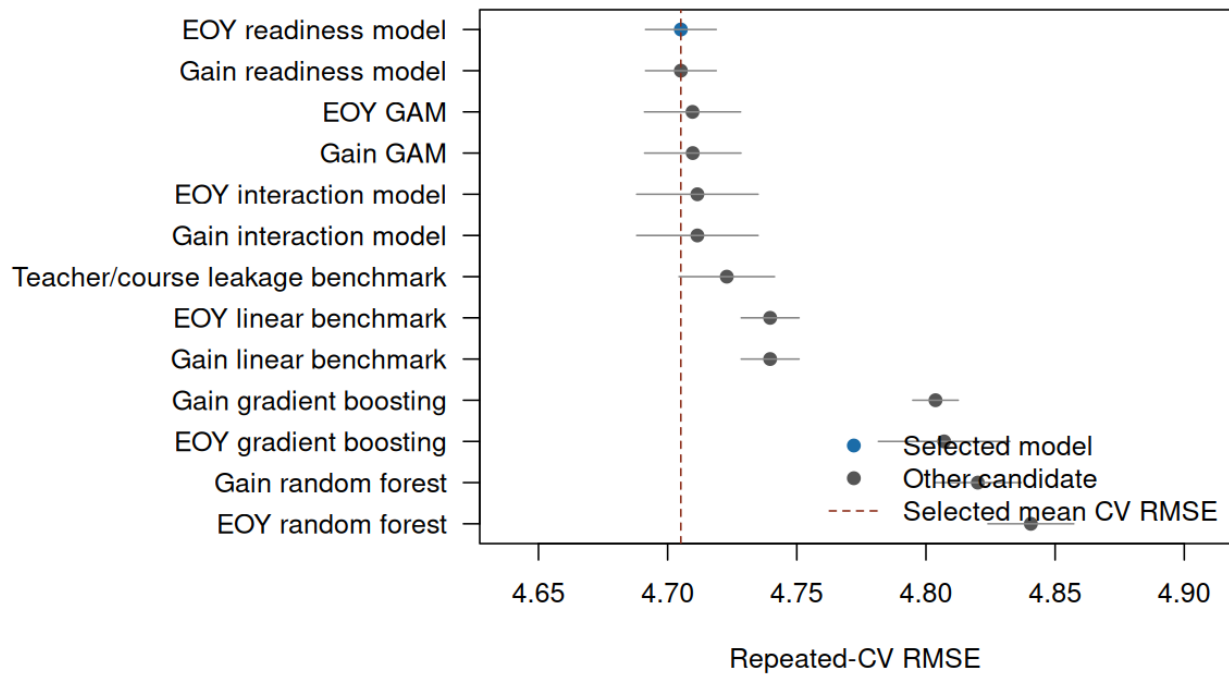
Model family review

Family	Representative model	Decision	CV RMSE	Holdout RMSE
Direct gain baseline	Gain readiness model	Compared	4.705	4.400
Predicted EOY baseline	EOY readiness model	Selected	4.705	4.400
Interaction surface	EOY interaction model	Compared	4.712	4.413
GAM smooths	EOY GAM	Compared	4.710	4.425
Random forest	Gain random forest	Compared	4.820	4.612
Gradient boosting	Gain gradient boosting	Compared	4.804	4.534
Teacher/course ID leakage check	Teacher/course leakage benchmark	Excluded	4.723	4.428

Model	Target	Method	CV RMSE	Gain R2	EOY R2	Holdout RMSE
EOY readiness	Predicted EOY	Linear model	4.705	0.158	0.937	4.400
Gain readiness	Predicted gain	Linear model	4.705	0.158	0.937	4.400
EOY GAM	Predicted EOY	GAM	4.710	0.156	0.937	4.425
Gain GAM	Predicted gain	GAM	4.710	0.156	0.937	4.425
EOY interaction	Predicted EOY	Linear model	4.712	0.155	0.937	4.413
Gain interaction	Predicted gain	Linear model	4.712	0.155	0.937	4.413
Leakage check	Predicted EOY	Linear model	4.723	0.151	0.937	4.428

The compact table shows the strongest candidates by repeated-CV RMSE plus the leakage check. The full model-comparison table is generated as [reports/growth_model_comparison_display.csv](#).

Expected-Growth Model Comparison

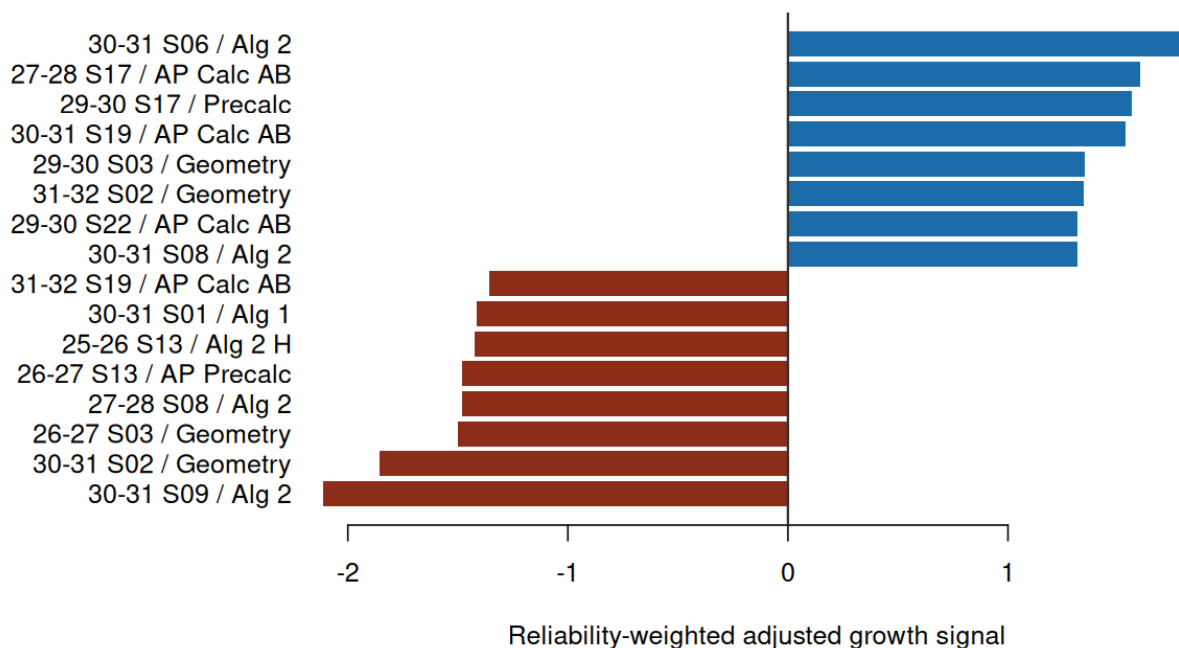


Historical Section Signals

For each historical section-year group, the adjusted signal is the reliability-weighted average residual: observed gain minus expected gain, weighted toward zero for smaller groups. Positive values mean the section improved more than expected for its starting mix; negative values mean it improved less than expected.

Section	Teacher	Course	N	Raw	Expected	Signal	Result
30-31 S06	TCH-002	Alg 2	9	10.39	6.53	1.83	Above
27-28 S17	TCH-005	AP Calc AB	11	8.79	5.74	1.60	In range
29-30 S17	TCH-003	Precalc	11	8.77	5.79	1.56	Above
30-31 S19	TCH-005	AP Calc AB	12	9.09	6.28	1.53	Above
26-27 S13	TCH-004	AP Precalc	10	2.27	5.23	-1.48	Below
25-26 S13	TCH-003	Alg 2 H	11	3.99	6.71	-1.42	In range
30-31 S01	TCH-001	Alg 1	10	3.28	6.10	-1.41	Below
31-32 S19	TCH-005	AP Calc AB	12	2.41	4.90	-1.36	In range

Sections Farthest Above or Below Expected Growth



The full section signal table is generated as [reports/section_adjusted_signals.csv](#) so reviewers can inspect all historical section-year groups, not only the highlights shown in the report.

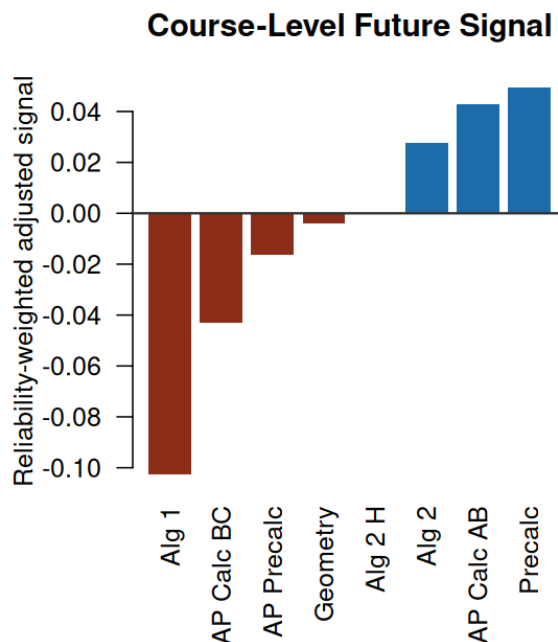
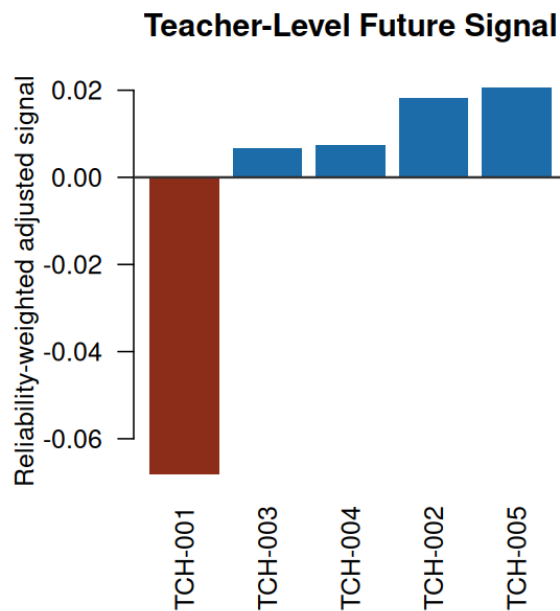
Teacher and Course Signal Summary

Teacher and course summaries aggregate the historical section evidence into future-facing review signals. They are useful for identifying where leaders may want to inspect pacing, curriculum alignment, attendance mix, or practices worth transferring. They should not be treated as standalone teacher quality scores.

Teacher	Sections	Records	Raw	Expected	Signal
TCH-005	41	420	4.81	4.77	0.02
TCH-002	40	424	6.48	6.45	0.02
TCH-004	36	313	5.11	5.10	0.01
TCH-003	34	351	5.95	5.93	0.01
TCH-001	23	229	6.49	6.67	-0.07

Course-level summary

Course	Sections	Records	Raw	Expected	Signal
Precalc	21	216	5.73	5.63	0.05
AP Calc AB	27	294	5.21	5.14	0.04
Alg 2	23	236	6.57	6.52	0.03
Alg 2 H	20	198	6.12	6.12	0.00
Beyond Core	6	21	3.08	3.08	0.00
Geometry	31	334	6.39	6.40	-0.00
AP Precalc	22	214	5.10	5.14	-0.02
AP Calc BC	14	131	3.86	3.97	-0.04
Alg 1	10	93	6.60	6.94	-0.10

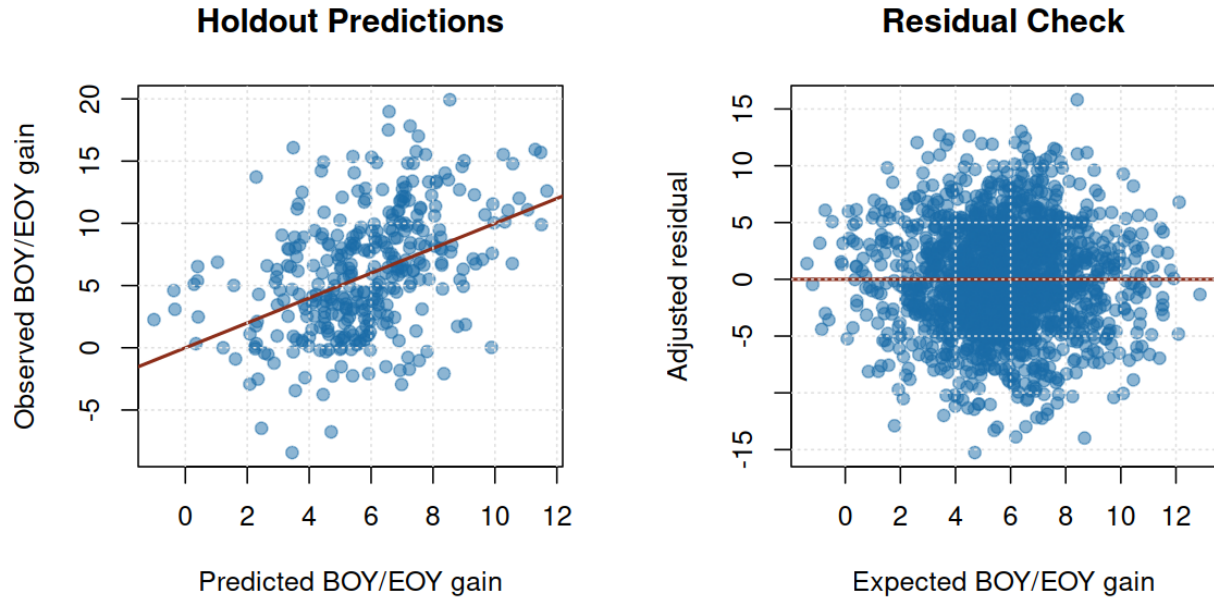


Diagnostics and Sensitivity

The diagnostics below separate two questions: how well the model predicts the expected baseline, and how much raw section rankings change after adjustment. The EOY R-squared describes baseline strength; the gain R-squared describes how noisy individual growth remains after BOY score is removed.

Diagnostic	Estimate	Interpretation
Holdout expected-gain RMSE	4.400	Typical holdout prediction error on BOY/EOY gain
Holdout expected-gain MAE	3.514	Average absolute holdout prediction error
Holdout expected-gain R-squared	0.180	Share of holdout gain variation explained by the expected-growth model
Holdout EOY RMSE	4.400	Typical holdout prediction error on final EOY score
Holdout EOY R-squared	0.947	Share of holdout EOY score variation explained by the baseline
Residual mean, all pairs	0.000	Near 0 means expected gain is centered overall
Residual SD, all pairs	4.598	Residual spread used to judge section signal uncertainty

Measure	Value
Included paired records	1,737
Section-year groups	174
Groups with at least 5 paired records	169
Groups with at least 8 paired records	159
Mean raw BOY/EOY gain	5.72
Raw-vs-adjusted rank correlation	0.797
Top-10 overlap, raw vs adjusted ranking	70.0%



Technical Appendix

The expanded model search used the installed R packages below. The operating model excludes persistent teacher, course, and section IDs; the leakage benchmark is reported only to show why those fields should not be used inside the baseline.

Package	Installed
mgcv	Available
randomForest	Available
gbm	Available

Metric	Value
Selected model	EOY readiness model
Selected target strategy	Predicted EOY
Selected method	Linear model
Candidate models tested	13
Operational candidates tested	12
Repeated CV folds	5
Repeated CV repeats	3
CV expected-gain RMSE	4.705
CV expected-gain R-squared	0.158
CV EOY R-squared	0.937
Holdout expected-gain RMSE	4.400
Holdout expected-gain R-squared	0.180
Holdout EOY R-squared	0.947
Mean raw BOY/EOY gain	5.72

The full selected-model metrics table is generated as [reports/growth_final_metrics.csv](#).

Bottom Line

- BOY/EOY improvement is the right stakeholder metric because it is easy to explain and aligned with instructional growth.
- The expected-score baseline should be used to create fairer teacher and course review signals, not to rank individual students or automate personnel decisions.
- Historical sections should be treated as evidence. Future action should focus on recurring teacher and course patterns before the next cycle.
- The model predicts final EOY score strongly, but individual gain remains noisy; aggregate signals and human review are necessary.

Reproducibility

Rebuild the full evidence packet with `make all`. The pipeline uses the included public-safe extract and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, real student-identifiable records, real personnel records, credentials, and copyrighted source documents.